

STA 2002

Lectures 18-19

Testing for homogeneity and independence

Chi-square test for homogeneity

Chi-square test for independence

Homogeneity and independence

A contingency table.

TABLE 11.2. Gender and Preference for Medical Care

	Type of medical care preference		
Gender	Western	Alternative	Total
Men	49 a	51 b	100 $(a + b)$
Women	30 c	70 d	100 $(c + d)$
Total	79 $(a + c)$	121 $(b + d)$	200 Grand total (n)

Note: Expected frequencies are shown in parentheses.

Do women and men behave differently in terms of medical care preference?

H_0: they behave the same.

What would the table look like if they indeed behave the same? Does the observed data strongly contradict with it?

Homogeneity and independence

A chi-square test for **homogeneity** is used in empirical investigations when the marginal totals for one condition have been fixed at certain values and the totals for the other condition may vary at random.

Ex:
Suppose that a researcher who is studying adult factory workers recruits 250 smokers and a comparison group of 300 nonsmokers. The researcher then refers the employees to a medical exam that assesses the presence of lung diseases; chronic cough is included in the review of symptoms.

TABLE 11.6. The Association between Smoking and Chronic Cough

	Diagnosis of chronic cough		
Smoking	Yes	No	Total
Yes	99	151	250
No	17	283	300
Total	116	434	550
			Grand total

Homogeneity and independence

Homogeneity:

We are comparing multiple distributions. The sampling can be done separately.

Ex:

Study question: Do Asians vote for Trump at a different rate than white?

Sampling scheme:

"Hey Tom, you go out and find 200 Asians and ask how they vote."

"Jerry, you go out and find 300 whites and ask how they vote."

These numbers (200 and 300) are usually pre-determined, and can be set almost independently.

There are budgeting/statistical power considerations that put some restraints/preferences on the numbers.

Homogeneity and independence

Thinking from the homogeneity perspective:

		Outcomes		
		Outcome 1	*Outcome 2*	
Variable X	*X=0*	*a*	*b*	*a+b*
	X=1	*c*	*d*	*c+d*
		a+c	*b+d*	*n*

What if the two row-wise distributions are the same?

$$P(Outcome\ 1 \mid X=0) = P(Outcome\ 1 \mid X=1) = P(Outcome\ 1)$$

Given the counts in the table,

$$\hat{P}(Outcome\ 1) = \frac{a+c}{n}$$

$$\hat{P}(Outcome\ 2) = \frac{b+d}{n}$$

$$\mathrm{E(a)} = (\mathrm{a} + \mathrm{b})\,\hat{P}(Outcome\ 1) = (a+b)\frac{(a+c)}{n}$$

$$\mathrm{E(b)} = (\mathrm{a} + \mathrm{b})\,\hat{P}(Outcome\ 2) = (a+b)\frac{(b+d)}{n}$$

$$\mathrm{E(c)} = (\mathrm{c} + \mathrm{d})\,\hat{P}(Outcome\ 1) = (c+d)\frac{(a+c)}{n}$$

$$\mathrm{E(d)} = (\mathrm{c} + \mathrm{d})\,\hat{P}(Outcome\ 2) = (c+d)\frac{(b+d)}{n}$$

Chi-square test on contingency table

TABLE 11.3. Notation Used in a 3 × 3 Contingency Table

	Variable Y			
Variable X	$n_{1,1}$	$n_{1,2}$	$n_{1,3}$	$n_{1.}$
	$n_{2,1}$	$n_{2,2}$	$n_{2,3}$	$n_{2.}$
	$n_{3,1}$	$n_{3,2}$	$n_{3,3}$	$n_{3.}$
	$n_{.1}$	$n_{.2}$	$n_{.3}$	Total = n

Now for each cell we have $O_{ij} = n_{i,j}$ and $E_{ij} = \frac{n_{i.}n_{.j}}{n}$, e.g., $E(n_{1,1}) = \frac{n_{1.}n_{.1}}{n}$

We can see this table as a multinomial distribution, with a number of extra constraints that reduce the *df*

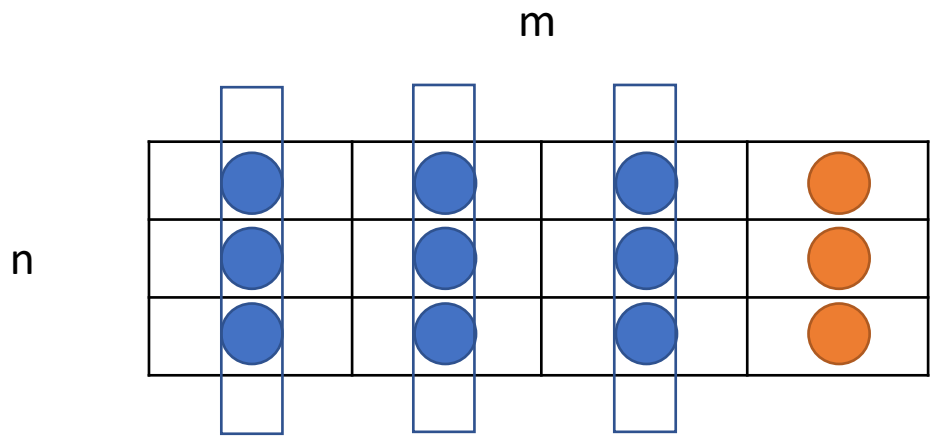
$$\chi^2 = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Chi-square test on contingency table

In contingency tables for homogeneity,

$$df = (n - 1)(m - 1)$$

n: # rows
m: #columns



- Total *nm* cells, with probability p_{ij} to occur for the (i, j)th cell.
- Lose *n* degrees of freedom to row constraints of $\sum_{j=1}^{m} p_{ij} = 1\,, \forall i$
- We need to estimate p_j by $\hat{p}_j = \frac{n_{\cdot j}}{n}$ for $j = 1, \ldots, (m - 1)$, so we lose *(m-1)* degrees of freedom to the estimation of these *m-1* parameters.
- $nm - n - (m - 1) = nm - n - m + 1 = (n - 1)(m - 1)$

Homogeneity and independence

A chi-square test for **independence** is used to assess whether two factors are related to each other.

Ex:

Some racial groups may tend to present with more advanced forms of breast cancer. Data from a hypothetical breast cancer staging study:

TABLE 11.4. Computation Example—the Association between Race/Ethnicity and Breast Cancer Stage in a Sample of Tumor Registry Patients

	Breast cancer stage			
Race	In situ	Local	Regional/distant	Total
White	124 (232.28)	761 (663.91)	669 (657.81)	1554
African American	36 (83.85)	224 (239.67)	301 (237.47)	561
Asian	221 (64.87)	104 (185.42)	109 (183.71)	434
Total	381	1089	1079	2549

Note: Expected values are shown in parentheses.

Homogeneity and independence

Independence:

We are considering **one joint distribution**. Ideally the samples should be collected as from a single distribution.

Ex:

Study question: In the city of Miami, do people who vote for Trump also tend to vote for DeSantis?

Sampling scheme:

"Hey Tom and Jerry, you guys go out and find 500 voters and ask how they vote."

Only the total number is pre-determined.

There are again budgeting/statistical power considerations that put some restraints/preferences on the number.

Homogeneity and independence

Thinking from the independence perspective:

		Variable Y		
		Y=0	Y=1	
Variable X	X=0	a	b	$a+b$
	X=1	c	d	$c+d$
		$a+c$	$b+d$	n

What if X and Y are independent from each other?

$$P(X=0,\ Y=0) = P(X=0)P(Y=0)$$

Given the counts in the table,

$$\hat{P}(X=0) = \frac{a+b}{n},\quad \hat{P}(Y=0) = \frac{a+c}{n},$$

$$\hat{P}(X=1) = \frac{c+d}{n},\quad \hat{P}(Y=1) = \frac{b+d}{n}$$

$$\mathrm{E(a)} = \mathrm{n}\,\hat{P}(X=0)\,\hat{P}(Y=0) = \frac{(a+b)(a+c)}{n}$$

$$\mathrm{E(b)} = \mathrm{n}\,\hat{P}(X=0)\,\hat{P}(Y=1) = \frac{(a+b)(b+d)}{n}$$

$$\mathrm{E(c)} = \mathrm{n}\,\hat{P}(X=1)\,\hat{P}(Y=0) = \frac{(c+d)(a+c)}{n}$$

$$\mathrm{E(d)} = \mathrm{n}\,\hat{P}(X=1)\,\hat{P}(Y=1) = \frac{(c+d)(b+d)}{n}$$

Chi-square test on contingency table

TABLE 11.3. Notation Used in a 3 × 3 Contingency Table

	Variable Y			
Variable X	$n_{1,1}$	$n_{1,2}$	$n_{1,3}$	$n_{1.}$
	$n_{2,1}$	$n_{2,2}$	$n_{2,3}$	$n_{2.}$
	$n_{3,1}$	$n_{3,2}$	$n_{3,3}$	$n_{3.}$
	$n_{.1}$	$n_{.2}$	$n_{.3}$	Total = n

Now for each cell we have $O_{ij} = n_{i,j}$ and $E_{ij} = \frac{n_{i.}n_{.j}}{n}$, e.g., $E(n_{1,1}) = \frac{n_{1.}n_{.1}}{n}$

We can see this table as a multinomial distribution, with a number of extra constraints that reduce the *df*

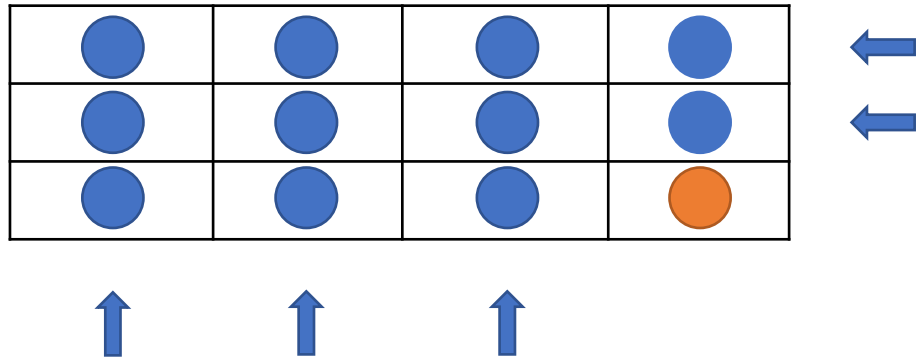
$$\chi^2 = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Chi-square test on contingency table

In contingency tables for independence,

$$df = (n - 1)(m - 1)$$

n: # rows
m: #columns



- Total *nm* cells.
- *1* degrees lost to probability constraint: $\sum_i \sum_j P(X, Y) = 1$
- Lose *(m-1)* degrees of freedom to the estimation of *m-1* parameters in the columns
- Lose *(n-1)* degrees of freedom to the estimation of *n-1* parameters in the rows
- $nm - 1 - (n - 1) - (m - 1) = nm - n - m + 1 = (n - 1)(m - 1)$

Homogeneity and independence

- A chi-square test for **homogeneity:**
  - We have several multinomial distributions with the same set of possible outcomes.
  - We want to test whether the cell probabilities of the multinomials are equal—that is, to test the homogeneity of the multinomial distributions.
  - Sampling is done independently for each multinomial distribution. The row margins are fixed.
- A chi-square test for **independence:**
  - We have a joint distribution.
  - We want to test whether two factors impact each other.
  - (Ideally) sampling is done as a single population. The grand total is fixed.
- The two tests use the ***exact same test statistic and the same degree of freedom***.
- Independence can be thought of as homogeneity of conditional distributions.

Examples for the test of homogeneity

H_0: There is no difference in preference for alternative medicine treatments between men and women.

TABLE 11.2. Gender and Preference for Medical Care

	Type of medical care preference		
Gender	Western	Alternative	Total
Men	49 a	51 b	100 $(a + b)$
Women	30 c	70 d	100 $(c + d)$
Total	79 $(a + c)$	121 $(b + d)$	200 Grand total (n)

Note: Expected frequencies are shown in parentheses.

$$\frac{(a+b)(a+c)}{n} = \frac{100 \times 79}{200} = 39.5 \qquad \mathrm{E}(n_{i,j}) = \frac{n_i . n_{.j}}{n} = \frac{row\ sum\ \times\ column\ sum}{grand\ total}$$

$$\frac{(49-39.5)^2}{39.5} + \frac{(30-39.5)^2}{39.5} + \frac{(51-60.5)^2}{60.5} + \frac{(70-60.5)^2}{60.5} = 7.55$$

$df = (n-1)(m-1) = 1$

$\chi^2_{1,0.05} = 3.84 < 7.55$

Null rejected, suggesting there is a difference.

Examples for the test of homogeneity

H_0: There is no association between smoking and chronic cough among adult factory workers.

TABLE 11.6. The Association between Smoking and Chronic Cough

	Diagnosis of chronic cough		
Smoking	Yes	No	Total
Yes	99 a	151 b	250
No	17 c	283 d	300
Total	116 $(a + c)$	434 $(b + d)$	550 Grand total (n)

$$\mathrm{E}(n_{i,j}) = \frac{row\ sum\ \times\ column\ sum}{grand\ total}$$

$$\mathrm{E}(n_{1,2}) = \frac{250 \times 434}{550} = 197.27$$

$$\frac{(99-52.73)^2}{52.73} + \frac{(17-63.27)^2}{63.27} + \frac{(151-197.27)^2}{197.27} + \frac{(283-236.73)^2}{236.73} = 94.33 \qquad df = 1$$

$\chi^2_{0.05}(1) = 3.84 < 94.33$

Null rejected, suggesting there is an association.

Examples for the test of homogeneity

Did the imitator do a good job?

Written by Jane Austen

Written by an admirer after Jane Austen died

Word	*Sense and Sensibility*	*Emma*	*Sanditon* I	*Sanditon* II
a	147	186	101	83
an	25	26	11	29
this	32	39	15	15
that	94	105	37	22
with	59	74	28	43
without	18	10	10	4
Total	375	440	202	196

Examples for the test of homogeneity

First, we consider Austen's consistency from one work to another.

Word	*Sense and Sensibility*	*Emma*	*Sanditon* I
a	147 160.0	186 187.8	101 86.2
an	25 22.9	26 26.8	11 12.3
this	32 31.7	39 37.2	15 17.1
that	94 87.0	105 102.1	37 46.9
with	59 59.4	74 69.7	28 32.0
without	18 14.0	10 16.4	10 7.5

$\chi^2 = 12.27$

df=10

$\chi^2_{0.1}(10) = 15.99 > 12.27$

Null is not rejected.

Examples for the test of homogeneity

```
> M=as.table(rbind(c(147,186,101),c(25,26,11),c(32,39,15),c(94,105,37),c(59,74,28),c(18,10,10)))
> dimnames(M) <- list(Word = c("a", "an", "this", "that", "with", "without"),party = c("Sense and Sensibility","Emma", "Sanditon I"))
> (Xsq=chisq.test(M))

        Pearson's Chi-squared test

data:  M
X-squared = 12.271, df = 10, p-value = 0.2673

> Xsq$expected
         party
Word      Sense and Sensibility      Emma Sanditon I
  a                   160.02950 187.76794  86.202557
  an                   22.86136  26.82399  12.314651
  this                 31.71091  37.20747  17.081613
  that                 87.02065 102.10423  46.875123
  with                 59.36578  69.65585  31.978368
  without              14.01180  16.44051   7.547689
```

Examples for the test of homogeneity

To compare Austen and her imitator, we can pool all Austen's work together in light of the above findings.

Word	Imitator	Austen
a	83 83.5	434 433.5
an	29 14.7	62 76.3
this	15 16.3	86 84.7
that	22 41.7	236 216.3
with	43 33.0	161 171.0
without	4 6.8	38 35.2

$\chi^2 = 32.81$

df=5

$\chi^2_{0.001}(5) = 20.52 < 32.81$

Null is rejected.

Examples for the test of Independence

Is there a relationship between marital status and educational level?

In a demographic study of women who were listed in *Who's Who*, Kiser and Schaefer (1949) compiled the following table for 1436 women who were married at least once:

Education	Married Once	Married More Than Once	Total
College	550	61	611
No College	681	144	825
Total	1231	205	1436

Education	Married Once	More Than Once
College	550 523.8	61 87.2
No College	681 707.2	144 117.8

The chi-square statistic is 16.01 with 1 degree of freedom, giving a *p*-value less than .001.

Independence can be thought of as homogeneity of conditional distributions. If education level and marital status are independent, then the conditional probabilities are homogeneous:

P(Married Once | College) = *P*(Married Once | No College).

Examples for the test of independence

TABLE 11.4. Computation Example—the Association between Race/Ethnicity and Breast Cancer Stage in a Sample of Tumor Registry Patients

	Breast cancer stage			
Race	In situ	Local	Regional/distant	Total
White	124	761	669	1554
African American	36 (	224	301	561
Asian	221	104	109	434
Total	381	1089	1079	2549

Note: Expected values are shown in parentheses.

$(1554 \times 381)/2549 = 232.2770$

chi-square statistic is 552.0993

df = (3 – 1)(3 – 1) = 4

$\chi^2_{0.001}(4) = 16.266 < 552.0993$

Null rejected, suggesting there is an association.

Examples for the test of independence

H_0: the numbers are a random sample

5.6	8.2	7.8	4.8	5.5	8.1	6.7	7.7	9.3	6.9
8.2	10.1	7.5	6.9	11.1	9.2	8.7	10.3	10.7	10.0
9.2	11.6	10.3	11.7	9.9	10.6	10.0	11.4	10.9	11.1

One way to reject the null is to ask: is there an upward trend?

Mark each x high (H) or low (L), depending on whether it is above or below the sample median.

Then divide the x values into three groups: $x_1,\ldots,x_{10}$; $x_{11},\ldots,x_{20}$; and $x_{21},\ldots,x_{30}$.

Group	L	H	Totals
1	9	1	10
2	5	5	10
3	1	9	10
Totals	15	15	30

Expected counts each cell $10 \times \frac{15}{30} = 5$

$df = (n-1)(m-1) = 2$

$$q = \frac{(9-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(9-5)^2}{5}$$

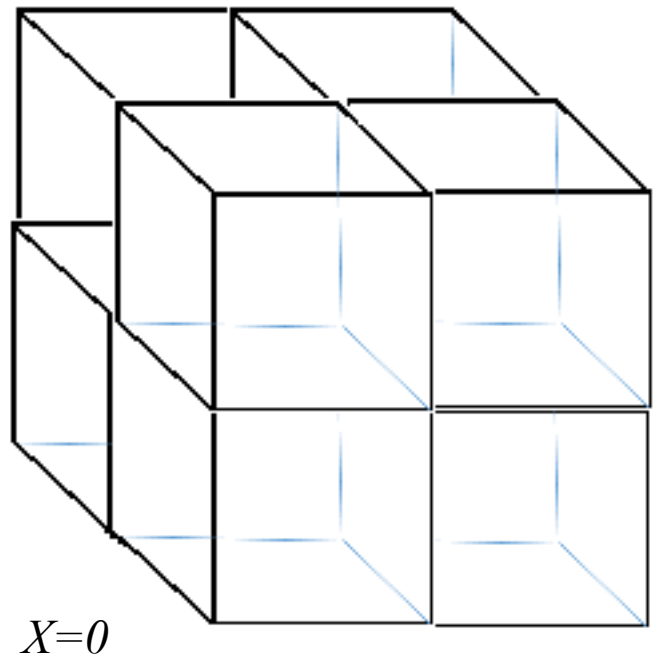
$$= 12.8 > 5.991 = \chi^2_{0.05}(2),$$

Alternatively, the p-value is

$$P(\chi^2(2) > 12.8) = 1 - pchisq(12.8,2) = 0.0017$$

Null is rejected. The numbers are not a random sample.

Three-way contingency table



H_0: *X, Y, Z are independent*

Under H_0,

$P(X=0, Y=0, Z=0) = P(X=0)P(Y=0)P(Z=0)$

$$\hat{P}(X = x_i) = \frac{n_{i..}}{n}, \qquad n_{i..} = \sum_j \sum_r n_{ijr}$$

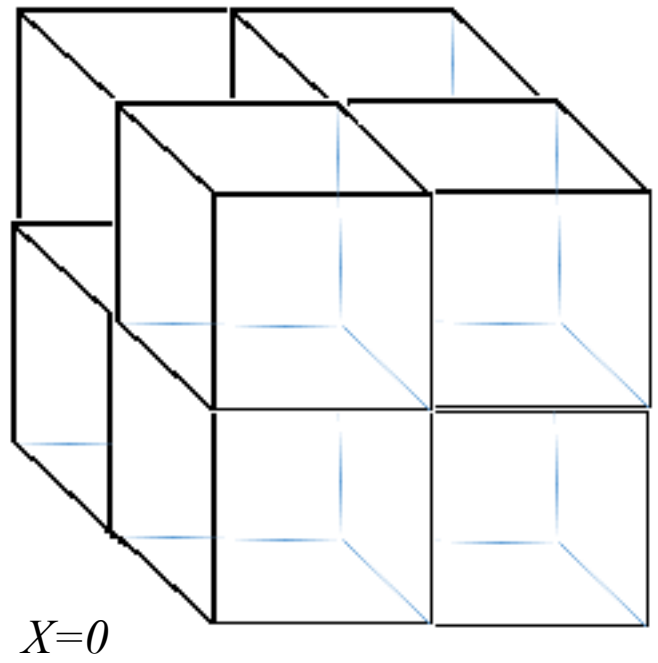
$$\hat{P}(Y = y_j) = \frac{n_{.j.}}{n}, \qquad n_{.j.} = \sum_i \sum_r n_{ijr}$$

$$\hat{P}(Z = z_r) = \frac{n_{..r}}{n}, \qquad n_{..r} = \sum_i \sum_j n_{ijr}$$

$$\hat{P}(X = x_i, Y = y_j, Z = z_r) = \hat{P}(X = x_i)\,\hat{P}(Y = y_j)\,\hat{P}(Z = z_r) = \frac{n_{i..}}{n}\frac{n_{.j.}}{n}\frac{n_{..r}}{n}$$

$$E(X = x_i, Y = y_j, Z = z_r) = n\frac{n_{i..}}{n}\frac{n_{.j.}}{n}\frac{n_{..r}}{n}$$

Three-way contingency table



$$E(X = x_i, Y = y_j, Z = z_r) = n\frac{n_{i..}}{n}\frac{n_{.j.}}{n}\frac{n_{..r}}{n}$$

The chi − *squared statistic*:

$$Q = \sum_r \sum_j \sum_i \frac{\left[n_{ijr} - n\frac{n_{i..}}{n}\frac{n_{.j.}}{n}\frac{n_{..r}}{n}\right]^2}{n\frac{n_{i..}}{n}\frac{n_{.j.}}{n}\frac{n_{..r}}{n}}$$

$$df = khm - 1 - (k-1) - (h-1) - (m-1) = khm - k - h - m + 2$$

total number of cells

Constraint $\sum_i \sum_j \sum_r P_{ijr} = 1$

Number of parameters estimated